

Super Mario Game — Global Specification

(FROZEN)

Version: 2.0 — 2026-06-02 **Status:** APPROVED — Do not modify without user confirmation.
Source: Compiled from user answers in [implementation_plan.md](#) Section 6, expanded with Feature Expansion Proposal. **Change Log:** v1.0 (2026-05-30) initial spec; v2.0 (2026-06-02) expanded from 51 → 110 features.

1. Project Overview

Field	Value
Project	CS202 Final Project — Super Mario Bros.
Language	C++17
Engine	SFML 3.0.2 + ImGui-SFML v3.0
Build	CMake with FetchContent
Rendering	2D only (no 3D)
Platform	macOS (primary), cross-platform via SFML
Total Features	110 (46 MVP + 59 Expanded + 5 Post-MVP Bonus)

2. Architecture

2.1 Layered Architecture

```
Application Layer    → main.cpp (entry point)
Core Layer           → Game, GameStateManager, InputManager, ResourceManager,
SoundManager, EventBus
Game States          → MenuState, CharSelectState, PlayingState, PauseState,
GameOverState,
                    VictoryState, OptionsState, WorldMapState, StatisticsState
World Layer          → World, TileMap, Camera, HUD, Minimap
Entity Layer         → Entity → Character → Player → Mario/Luigi/Toad/Peach; Entity →
Character → Enemy; Entity → Item; Entity → Block
Physics Layer        → PhysicsEngine, AABB, CollisionDetector, CollisionResolver,
SpatialHash
Infrastructure        → Renderer, AnimationManager, Serializer, EntityFactory, EventBus,
ObjectPool, ReplayRecorder, AchievementManager,
StatisticsTracker
```

2.2 Design Patterns (10+ Total — 5 Required + 5+ Bonus)

#	Pattern	Class(es)	Purpose
1	Factory	<code>EntityFactory</code>	<code>createEntity(type, pos)</code> → <code>std::unique_ptr<Entity></code> . Spawns all 25+ entity types. Lakitu uses Factory to spawn Spinies.
2	Singleton	<code>Game</code> , <code>ResourceManager</code> , <code>SoundManager</code> , <code>AchievementManager</code>	Single instances. Lazy-init via <code>getInstance()</code> .
3	State	<code>GameStateManager</code> / <code>IGameState</code> , <code>IPlayerState</code> , <code>FallingPlatform</code> , <code>Thwomp</code>	Game flow (9 states). Player forms (5 concrete base states). Entity lifecycles.
4	Observer	<code>EventBus</code>	Publish/subscribe: 15+ event types. HUD, SoundManager, ComboTracker, AchievementTracker, StatisticsTracker subscribe.
5	Strategy	<code>IMovementStrategy</code>	Enemy AI: 7+ strategies (Patrol, Chase, Fly, Swim, TetheredChase, HammerThrow, TimerEmergence). Swappable at runtime. <code>DifficultyStrategy</code> for game-wide scaling.
6	Command	<code> ICommand</code> + <code>InputManager</code>	8+ commands. Keyboard → game commands. Debug console text→command parsing. Replay serialization.
7	Decorator	<code>StarDecorator</code> , <code>MegaDecorator</code>	Temporary state overlays that wrap the active <code>IPlayerState</code> without modifying it.
8	Memento	<code>GameSnapshot</code>	Time rewind stores/restores full game state snapshots. Replay system.
9	Object Pool	<code>ObjectPool<T></code>	Pre-allocated pools for fireballs, particles, Bullet Bills. Avoids allocation in hot loops.
10	Template Method	<code>IMovementStrategy::execute()</code>	Base <code>execute()</code> defines skeleton: <code>calculateTarget()</code> → <code>applyMovement()</code> → <code>checkConstraints()</code> . Concrete strategies override hooks.

3. Window & Rendering

Setting	Value
Window Resolution	1280 × 720 pixels
Tile Size	32 × 32 pixels
Visible Width	40 tiles (1280 / 32)
Visible Height	22.5 tiles (720 / 32)
Art Style	16-bit SNES style
Sprite Source	Open-source (OpenGameArt, The Spriters Resource, etc.)
Sprite Dimensions	32 × 32 base
Scrolling	Horizontal (primary), vertical (tower/beanstalk sections)
Background	Parallax scrolling (multiple layers: sky, clouds, mountains, ground)
Particles	Brick fragments, coin sparkles, death poof, stomp effects, combo stars
Transitions	Fade-in/fade-out between levels and screens
Dev Tools	ImGui-SFML overlay (toggle with F12 or similar)
Screen Shake	Configurable on: block break, ground pound, Bowser stomp, Mega Mushroom

4. Physics & Gameplay Constants

All values are exposed to ImGui for real-time tuning.

Constant	Initial Value	Notes
Gravity	0.5 px/frame ² (= 1800 px/s ² at 60fps)	Tunable via ImGui
Mario Walk Speed	150 px/s	Tunable via ImGui
Mario Run Speed	300 px/s	Tunable via ImGui
Mario Jump Height	~4 tiles (128 px at 32px/tile)	Tunable via ImGui
Fixed Timestep	1/60 s (16.667 ms)	With interpolated rendering
Physics	Custom AABB + Spatial Hash	Broad-phase grid for O(n) collision
Coyote Time	6 frames (100ms)	Grace period after leaving ledge
Jump Buffer	6 frames (100ms)	Input buffer before landing

4.1 Luigi Modifiers (Relative to Mario)

Attribute	Multiplier
Jump Force	×1.2 (jumps higher)
Walk/Run Speed	×0.85 (slower)
Airborne Gravity	×0.9 (floatier)
Special	Double jump ability

4.2 Fireball Mechanics

Property	Value
Max Active	2 at once
Behavior	Bounces on ground
Destruction	On wall impact
Fire Rate	~0.3s cooldown between shots

4.3 Advanced Movement Mechanics [NEW in v2.0]

4.3.1 Wall Jump / Wall Slide

Property	Value
Wall Slide Speed	50 px/s (capped downward velocity)
Wall Jump Force	Same as regular jump, angled 45° away from wall
Wall Detect	Must hold direction toward wall while airborne
Animation	New <code>wall_slide</code> sprite, dust particles on jump

4.3.2 Ground Pound

Property	Value
Trigger	Press Down while airborne
Hover Time	0.15s pause at apex before slamming
Slam Speed	600 px/s (instant downward)
Effects	Breaks bricks below, stuns nearby enemies (2-tile radius), screen shake
Damage	Kills stompable enemies on impact

4.3.3 Crouch / Slide

Property	Value
Crouch	Hold Down on ground → reduce hitbox to 1 tile height (from 2 when Super+)
Slide	Hold Down while running → slide for 0.5s, pass under 1-tile gaps
Slide Damage	Kills enemies on contact during slide

4.3.4 Swimming / Water Sections

Property	Value
Gravity	×0.3 (reduced in water)
Swim Button	Jump button = swim stroke (upward impulse)
Max Sink Speed	60 px/s
Horizontal Speed	×0.6 of ground speed
Fireballs	Travel straight in water (no bounce)
Visual	Blue overlay tint, bubble particles

4.3.5 Climbing (Vines / Beanstalks)

Property	Value
Climb Speed	100 px/s
Trigger	Press Up when overlapping a vine/beanstalk tile
Controls	Up/Down to climb, Jump to leap off, Left/Right to dismount
Destination	Leads to bonus cloud areas above the main level

4.3.6 Damage Knockback

Property	Value
Knockback Force	150 px/s horizontal, 100 px/s vertical
Stun Duration	0.3s (player cannot input during stun)
Direction	Away from damage source

4.3.7 Momentum & Acceleration Curves

Property	Value
Ground Acceleration	0 → walk speed in 0.2s (non-linear ease-in)
Ground Deceleration	Walk speed → 0 in 0.15s
Air Acceleration	×0.5 of ground acceleration
Air Friction	×0.3 of ground friction
Skid Trigger	Reversing direction while moving above 100 px/s
Skid Animation	Dust particles + skid sprite

4.3.8 Combo System

Property	Value
Trigger	Consecutive enemy stomps without touching ground
Multiplier	×1, ×2, ×4, ×8 (caps at ×8)
Score	Base stomp score × multiplier
Display	Combo counter on HUD, escalating SFX pitch
Reset	Touching ground or taking damage

5. Characters

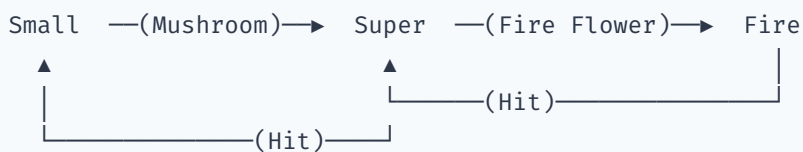
5.1 Player Characters

Character	Select	Special
Mario	Default	Standard physics
Luigi	Alt	Higher jump, slower, floatier, double jump
Toad	Unlockable (complete all 3 levels)	×1.3 speed, ×0.8 jump, no slide friction delay
Peach	Unlockable (complete all 3 levels with no deaths)	Float ability (hold jump to hover for 1.5s), ×0.9 speed

- Switching: Both at character select screen AND during gameplay via hotkey.
- Two-player: Same keyboard, versus mode.

5.2 Player States (State & Decorator Patterns)

Base States (IPlayerState)



Super -- "(Cape Feather)" --> Cape

Small -- "(Mini Mushroom)" --> Mini

Decorators (Wrap Current State)

- StarDecorator = 10s invincible overlay
- MegaDecorator = 8s giant overlay

State / Decorator	Size	Abilities	On Hit
Small	1 tile (32px)	Walk, run, jump	Die (lose life)
Super	2 tiles (64px)	Walk, run, jump, break bricks	→ Small (invincibility frames)
Fire	2 tiles (64px)	Walk, run, jump, break bricks, shoot fireballs	→ Super (invincibility frames)
Cape	2 tiles (64px)	Walk, run, jump, break bricks, glide, swoop attack	→ Super (invincibility frames)
Mini	0.5 tiles (16px)	Walk on water, enter mini pipes, floatier jump	Die (lose life — fragile)
(Dec) Mega	4 tiles (128px)	Destroys everything on contact. Temporary (8s).	Cannot be hit (invincible during Mega)
(Dec) Star	Same as current	All current + invincible, kills enemies on contact	Timer expires → return to base state

6. Enemies

6.1 Enemy Roster

Enemy	AI Strategy	Stomp Result	Fireball Result	Points
Goomba	PatrolStrategy — walks one direction, reverses on wall, falls off ledges	Dies (squish animation)	Dies	100
Red Goomba	PatrolStrategy — walks, reverses on wall AND at ledges (won't fall)	Dies (squish animation)	Dies	100
Koopa Troopa	PatrolStrategy — walks, reverses on wall	Retreats into shell (kickable)	Dies	200
Red Koopa	PatrolStrategy — reverses at ledges (ledge-aware)	Retreats into shell	Dies	200
Koopa Paratroopa	FlyStrategy — horizontal patrol + sinusoidal vertical	Loses wings → becomes Koopa Troopa	Dies	400
Red Paratroopa	FlyStrategy — stationary, vertical bounce only	Loses wings → becomes Red Koopa	Dies	400
Boo (Ghost)	ChaseStrategy — idle until player within 250px, chases when back turned	Cannot be stomped	Cannot be killed by fireball	N/A
Piranha Plant	TimerEmergenceStrategy — emerges from pipe on 3s timer, retreats for 2s	Cannot be stomped	Dies	100
Bullet Bill	LinearStrategy — fired from Bill Blaster, travels horizontally at 200 px/s	Dies (stompable)	Immune	200
Hammer Bro	HammerThrowStrategy — jumps between platforms, throws hammers in arcs	Dies	Dies	1000
Thwomp	ProximityTriggerStrategy — idle on ceiling, slams down when player within 3 tiles below	Cannot be stomped (damages player)	Immune	N/A
Chain Chomp	TetheredChaseStrategy — lunges toward player but snaps back to 4-tile radius	Cannot be stomped	Immune	N/A
Lakitu	FlyStrategy — hovers above, follows player, drops Spiny eggs every 4s	Dies (stompable when below)	Dies	800

Enemy	AI Strategy	Stomp Result	Fireball Result	Points
Spiny	<code>PatrolStrategy</code> — walks on ground after hatching from egg	Cannot be stomped (damages player)	Dies	100
Boom Boom (Mid-Boss)	Custom — charges and spins, pauses to recover. 3 stomps to defeat.	Takes 1 hit (3 total needed)	Takes damage	2000
Bowser (Boss)	Custom boss AI — Phase 1 (walk + breathe fire), Phase 2 (jump + faster fire)	Multi-hit (health bar)	Takes damage	5000

6.2 Proximity AI Detail (Boo)

- **Idle Range:** > 250 pixels from player
- **Chase Range:** ≤ 250 pixels from player
- **Behavior:** Moves toward Mario when Mario's back is turned. Freezes when Mario faces it (classic Boo behavior).
- **Invulnerable:** Cannot be stomped or killed by fireballs. Player must avoid.

6.3 Enemy Variants [NEW in v2.0]

Color variants alter enemy behavior without creating new classes:

Base Enemy	Variant	Behavioral Difference
Goomba (brown)	Red Goomba	Ledge-aware (turns at edges instead of falling)
Koopa (green)	Red Koopa	Ledge-aware (turns at edges)
Paratroopa (green)	Red Paratroopa	Vertical-only bounce (no horizontal patrol)

Variants are configured via JSON: `{ "type": "goomba", "variant": "red" }` → Factory applies different Strategy.

6.4 Mid-Boss: Boom Boom [NEW in v2.0]

- **Location:** End of Level 2 (Underground)
- **Phases:**
 - Hit 1: Normal charge speed
 - Hit 2: Faster charge + shorter recovery

- **Hit 3:** Fastest charge + spin attack
- **Arena:** Enclosed room, no escape until defeated
- **Reward:** Opens path to Level 2 flagpole

7. Items & Power-ups

Item	Source	Effect	Points
Super Mushroom	QuestionBlock	Small → Super	1000
Fire Flower	QuestionBlock	Super → Fire (if Small, acts as Mushroom first)	1000
Coin	QuestionBlock, floating in level	+1 coin, +200 score. 100 coins = 1 extra life	200
Star	QuestionBlock	10 seconds invincibility overlay	1000
1-UP Mushroom	QuestionBlock (rare)	+1 life	—
Cape Feather	QuestionBlock	Super → Cape (glide + swoop attack)	1000
Mega Mushroom	QuestionBlock (very rare)	Temporary giant form (8s). Destroys everything.	1000
Mini Mushroom	QuestionBlock (rare)	Shrinks to half-tile. Walk on water, enter mini pipes.	1000
POW Block	Placed in level (not from QuestionBlock)	Hit/throw → screen shake, all grounded enemies die/flip	—
P-Switch	BrickBlock (rare)	Bricks ↔ Coins for 15 seconds. Timer countdown on HUD.	—
Trampoline	Placed in level	Bounces player ~6 tiles high. Can be carried and placed.	—
Star Coin	Placed in level (3 per level)	Collectible. Tracked per level. Unlocks bonus content.	1000

- Items spawn from QuestionBlocks and hidden blocks.
- Power-down chain: Fire/Cape → Super → Small (step-down on each hit).

- Mega and Star are temporary overlays that expire.

8. Blocks & World Objects

Block	Behavior	Breakable?
BrickBlock	Hit from below. Breaks if Super/Fire/Cape. Contains coins sometimes. Particle effect on break.	Yes (Super+)
QuestionBlock	Hit from below → spawns contained item. Becomes empty block. Bump animation.	No
Hidden Block	Invisible until hit from below. Can contain 1-UP or coins.	No
Pipe	Warp to bonus areas or between level sections. Enter with Down. Piranha Plants emerge from some.	No
Flagpole	Level end marker. Player slides down. Score bonus based on height caught.	No
Moving Platform	Travels on a fixed path (horizontal or vertical). Player rides on top. Carries entities.	No
Falling Platform	Stable → Shaking (player stands for 1s) → Falling → Respawns after 5s.	No
Ice Block	Surface with reduced friction. Player slides further. Affects enemies too.	No
Conveyor Belt	Pushes entities in indicated direction. Visual arrow animation. Speed: 100 px/s.	No

8.1 Moving Platform Specifications [NEW in v2.0]

Property	Value
Speed	60 px/s
Path Types	Horizontal (back-and-forth), Vertical (back-and-forth), Circular
Player Interaction	Player inherits platform velocity when standing on it
JSON Config	<pre>{ "type": "moving_platform", "path": [{ "x": 10, "y": 15 }, { "x": 20, "y": 15 }], "speed": 60 }</pre>

8.2 Falling Platform Specifications [NEW in v2.0]

State	Duration	Behavior
Stable	Until player stands	Normal solid block
Shaking	1 second	Visual shake, warning to player
Falling	Until off-screen	Falls with gravity, passes through other tiles
Respawning	5 seconds	Invisible, then reappears at original position

9. Levels & World

9.1 Level Specifications

Level	Theme	Width	Difficulty	Unique Features
Level 1	Overworld / Grassland	200 tiles (6400px)	Easy	Few enemies, flat terrain, tutorial-like, swimming section
Level 2	Underground / Cave	200 tiles (6400px)	Medium	Gaps/pits, more enemies, darker palette, Boom Boom mid-boss, ice blocks
Level 3	Castle / Lava	200 tiles (6400px)	Hard	Lava pits, high enemy density, Thwomps, autoscroll section, Bowser boss

9.2 Level Structure

Property	Value
Format	3 continuous stages, each with mid-level checkpoints
File Format	JSON
Height	22.5 tiles visible (720px / 32px)
End Marker	Flagpole (classic Mario)
Warp Pipes	Yes — teleport to bonus coin areas or between level sections
Checkpoints	Mid-level checkpoint flags. On death: respawn at last checkpoint.
Star Coins	3 per level, placed in challenging-to-reach locations

9.3 World Map [NEW in v2.0]

Property	Value
Style	Super Mario Bros. 3 / World style overhead map
Nodes	Level 1, Level 2, Level 3, Bonus stages
Navigation	Arrow keys to move between nodes, Enter to start level
Display	Star coin count per level, completion status (star/flag icon)
State	<code>WorldMapState</code> in GameStateManager
Unlock	Levels unlock sequentially (must complete Level N to access Level N+1)

9.4 Bonus Rooms [NEW in v2.0]

Property	Value
Access	Via hidden warp pipes in main levels
Type	Coin-filled rooms with a 15-second timer
Reward	Collect as many coins as possible before timer expires
Exit	Automatic warp back to main level after timer

9.5 Special Level Sections [NEW in v2.0]

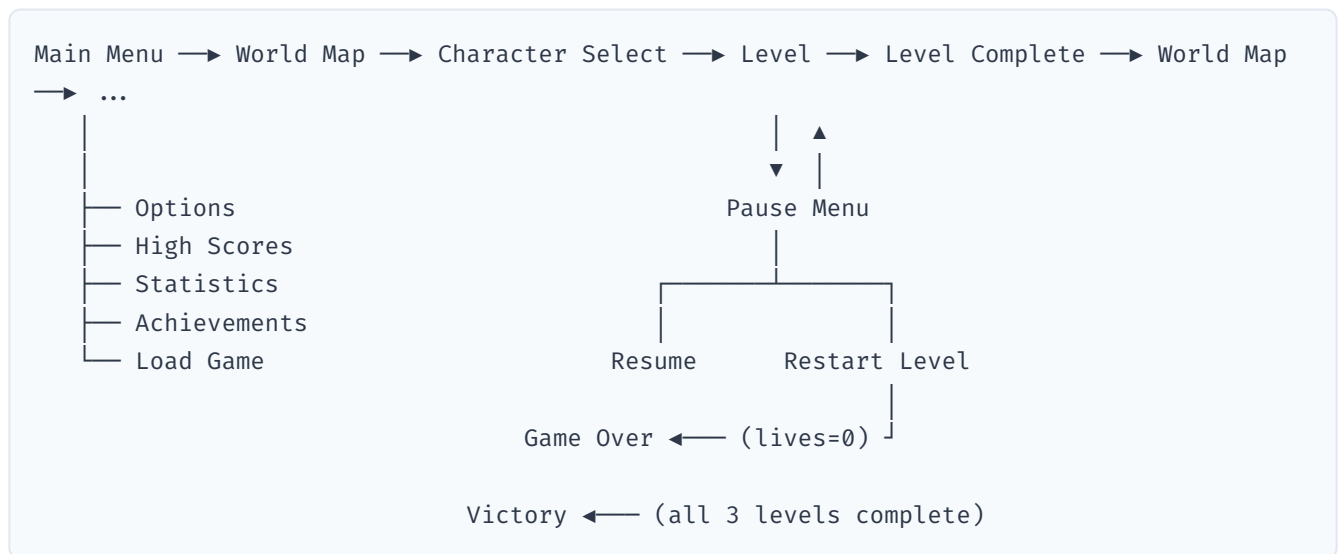
Section Type	Description	Camera Behavior
Vertical Scroll	Tower/beanstalk sections — player climbs upward	Camera follows vertically
Autoscroll	Airship-style — camera moves right automatically at 80 px/s	Player dies if pushed off left edge
Water Section	Submerged area within a level — different physics apply	Normal horizontal scroll

9.6 Level JSON Schema

```
{
  "name": "Level 1 - Grassland",
  "theme": "overworld",
  "width": 200,
  "height": 23,
  "tileSize": 32,
  "backgroundLayers": ["sky.png", "clouds.png", "mountains.png"],
  "spawnPoint": { "x": 2, "y": 20 },
  "checkpoints": [{ "x": 100, "y": 20 }],
  "flagpole": { "x": 198, "y": 8 },
  "starCoins": [
    { "x": 45, "y": 10 },
    { "x": 120, "y": 5 },
    { "x": 180, "y": 12 }
  ],
  "waterZones": [
    { "x1": 60, "y1": 15, "x2": 80, "y2": 22 }
  ],
  "tiles": [
    { "type": "ground", "x": 0, "y": 22, "w": 50 },
    { "type": "brick", "x": 20, "y": 16 },
    { "type": "question_block", "x": 22, "y": 16, "item": "coin" },
    { "type": "hidden_block", "x": 30, "y": 14, "item": "oneup" },
    { "type": "ice", "x": 70, "y": 22, "w": 10 },
    { "type": "conveyor", "x": 90, "y": 22, "w": 5, "direction": "right" }
  ],
  "entities": [
    { "type": "goomba", "x": 30, "y": 21 },
    { "type": "goomba", "x": 35, "y": 21, "variant": "red" },
    { "type": "koopas", "x": 50, "y": 21 },
    { "type": "piranha_plant", "x": 40, "y": 20, "pipe_id": 1 },
    { "type": "bullet_blaster", "x": 85, "y": 18 },
    { "type": "coin", "x": 25, "y": 14 }
  ],
  "movingPlatforms": [
    { "path": [{ "x": 60, "y": 15 }, { "x": 75, "y": 15 }], "speed": 60 }
  ],
  "pipes": [
    { "id": 1, "entrance": { "x": 40, "y": 20 }, "exit": { "x": 40, "y": 10 }, "target":
"bonus_area_1" }
  ]
}
```

10. Game Flow & UI

10.1 Screen Flow



10.2 Main Menu Options

- New Game
- Load Game
- Options (Volume: SFX slider + Music slider, Difficulty, Controls)
- High Scores (persisted across sessions)
- Statistics
- Achievements
- Quit

Animated Main Menu: Mario runs across a parallax background. Animated title logo. Coins spin. If idle for 30 seconds, demo playback starts (attract mode).

10.3 Pause Menu Options

- Resume
- Restart Level
- Save Game (manual save to slot)
- Return to World Map
- Return to Main Menu
- Quit

10.4 HUD Elements (Always Visible During Gameplay)

Element	Position	Format
Score	Top-left	SCORE 000000
Coins	Top-center-left	×00 with coin icon
World/Level	Top-center	WORLD 1-1
Time	Top-center-right	TIME 300 (countdown)
Lives	Top-right	×3 with character icon
Combo	Center (temporary)	×2! , ×4! , ×8! with escalating size
P-Switch Timer	Top (when active)	P-SWITCH 12 countdown bar
Boss Health	Top-center (boss fights)	Health bar with boss name
Star Coins	Below time	3 coin outlines, filled when collected

10.5 Floating Score Text [NEW in v2.0]

- "+200", "+1UP", "×4 COMBO!" text floats upward from event location and fades over 1 second.
- Uses particle-like text objects managed by HUD.

10.6 Minimap [NEW in v2.0]

Property	Value
Position	Bottom-right corner
Size	200 × 40 pixels
Shows	Full level overview, player dot (green), enemy dots (red), item dots (yellow)
Toggle	Press M to show/hide
Opacity	60% transparent

10.7 Statistics Screen [NEW in v2.0]

Tracks across all play sessions: - Total enemies defeated (by type) - Total coins collected - Total deaths - Total time played - Levels completed - Highest combo achieved - Star coins collected (per level)

10.8 Controls Rebinding [NEW in v2.0]

- Available in Options menu.
- Each action can be remapped to any key.
- Persisted to `config.json`.
- Shows current key assignments with visual keyboard.
- Conflict detection (warns if two actions share a key).

10.9 Death Counter / Retry Screen [NEW in v2.0]

- Shows per-level death count on Game Over.
- Encouraging messages cycle: "Try again!", "You've got this!", "Almost there!"
- Quick Retry button skips menu navigation.

10.10 Lives & Game Over

- Start with **3 lives** (Easy: 5 lives, Hard: 1 life).
 - On death: respawn at **last checkpoint** (or level start if no checkpoint reached).
 - On Game Over (lives = 0): return to **World Map** (or Main Menu if no progress).
 - Time limit: **300 game-seconds** per level.
-

11. Audio

11.1 Sound Effects (SFX)

Event	SFX File
Jump	<code>sfx/jump.wav</code>
Coin Collect	<code>sfx/coin.wav</code>
Stomp Enemy	<code>sfx/stomp.wav</code>
Power-up	<code>sfx/powerup.wav</code>
Power-down	<code>sfx/powerdown.wav</code>
Fireball	<code>sfx/fireball.wav</code>
1-UP	<code>sfx/oneup.wav</code>
Player Death	<code>sfx/death.wav</code>
Flagpole	<code>sfx/flagpole.wav</code>
Pipe Warp	<code>sfx/pipe.wav</code>
Ground Pound	<code>sfx/groundpound.wav</code>
Wall Jump	<code>sfx/walljump.wav</code>
Combo (escalating)	<code>sfx/combo_1.wav</code> → <code>sfx/combo_4.wav</code>
P-Switch	<code>sfx/pswitch.wav</code>
POW Block	<code>sfx/pow.wav</code>
Achievement Unlock	<code>sfx/achievement.wav</code>
Splash (water)	<code>sfx/splash.wav</code>

11.2 Background Music (BGM)

Context	BGM File
Main Menu	<code>music/menu.ogg</code>
World Map	<code>music/worldmap.ogg</code>
Overworld (Level 1)	<code>music/overworld.ogg</code>
Underground (Level 2)	<code>music/underground.ogg</code>
Castle (Level 3)	<code>music/castle.ogg</code>
Star Power	<code>music/star.ogg</code> (overrides level BGM while active)
Boss Fight	<code>music/boss.ogg</code>
Bonus Room	<code>music/bonus.ogg</code>
Game Over	<code>music/gameover.ogg</code> (jingle)
Victory	<code>music/victory.ogg</code> (jingle)

11.3 Audio Controls

- Separate volume sliders: SFX and Music (in Options menu).
- Source: Open-source / royalty-free retro-style audio.

11.4 Surface-Dependent Footsteps [NEW in v2.0]

Surface	Sound
Grass/Ground	Soft footstep
Stone/Brick	Hard footstep
Ice	Sliding/scraping
Metal (Castle)	Metallic clang

11.5 Dynamic Music Layers [NEW in v2.0]

- Base layer always plays.
- Additional instrument layers fade in based on:
- Proximity to boss (drums intensify)

- Low time remaining (tempo increases)
 - Star power (override to star track)
 - Combo streak (percussion layer adds)
-

12. Save/Load (Serialization)

12.1 Format

JSON — human-readable, easy to debug.

12.2 Persisted Data

```
{
  "version": "2.0",
  "timestamp": "2026-06-02T14:30:00Z",
  "player": {
    "character": "mario",
    "state": "fire",
    "position": { "x": 1024.5, "y": 640.0 },
    "lives": 3,
    "score": 12500,
    "coins": 47
  },
  "level": {
    "id": 2,
    "name": "Underground",
    "timeRemaining": 185.5,
    "checkpoint": { "x": 100, "y": 20 },
    "starCoinsCollected": [true, false, false]
  },
  "progress": {
    "levelsCompleted": [1],
    "unlockedCharacters": ["mario", "luigi"],
    "starCoins": { "1": [true, true, false], "2": [true, false, false], "3": [false,
false, false] }
  },
  "statistics": {
    "totalEnemiesDefeated": 142,
    "totalCoinsCollected": 523,
    "totalDeaths": 18,
    "totalTimePlayed": 3600.5,
    "highestCombo": 6
  },
  "achievements": ["first_stomp", "100_coins", "beat_bowser"],
  "highScores": [50000, 42000, 35000, 28000, 12500],
  "settings": {
    "sfxVolume": 80,
    "musicVolume": 60,
    "difficulty": "normal",
    "keyBindings": { "jump": "W", "left": "A", "right": "D", "fire": "F" }
  }
}
```

12.3 Save Slots

- 3 save slots available.
- Auto-save at checkpoint.

- Manual save from pause menu.
- Save slot preview shows: character, level, score, star coins, play time.

13. Two-Player Mode (Versus)

Setting	Value
Mode	Same keyboard, versus
Player 1	WASD + Q (fire) + E (special)
Player 2	Arrow keys + M (fire) + N (special)
Objective	Both players in same level. Race to flagpole. Can stomp each other (or configurable).

14. Controls

14.1 Single-Player Keyboard Mapping (Default)

Action	Key
Move Left	A or ←
Move Right	D or →
Jump	W or ↑ or Space
Run	Left Shift (hold)
Crouch / Ground Pound	S or ↓
Fire	F or J
Pause	Escape
Switch Character	Tab
Minimap Toggle	M
Dev Tools (ImGui)	F12
Debug Console	~ (tilde)

14.2 Input Architecture

- **Command Pattern:** Keyboard → `ICommand` objects → Character actions.
- Abstraction layer for future gamepad/controller support.
- All keys rebindable from Options menu.
- Key bindings persisted to `config.json`.

15. Visual Effects [NEW in v2.0]

15.1 Screen Shake System

Trigger	Intensity	Duration
Brick break	Light (2px offset)	0.1s
Ground pound	Medium (4px offset)	0.2s
Bowser stomp	Heavy (6px offset)	0.3s
Mega Mushroom movement	Continuous light	While active
POW Block	Heavy (6px offset)	0.3s

15.2 Entity Death Animations

Entity	Death Animation
Goomba	Squish flat for 0.5s, then disappear
Koopa (fireball)	Flip upside-down, fall off screen
All enemies (Star kill)	Launch into air spinning
Player	Jump animation upward, then fall off screen

15.3 Invincibility Visual FX

Type	Effect
Star Power	Rainbow color cycling on player sprite (6 colors, 0.1s per color) + sparkle trail particles
Hit Invincibility	Sprite flashes (visible/invisible alternating, 0.1s interval) for 2 seconds

15.4 Water / Lava Animation

- Water surface: sine-wave vertex displacement, blue tint overlay, bubble particles rising.
- Lava surface: orange/red animated tiles, ember particles, heat distortion shader (optional).

16. Advanced Systems [NEW in v2.0]

16.1 Object Pooling

Pool	Pre-allocated Size	Entity
Fireballs	10	Player fireballs + Bowser fireballs
Particles	200	All particle types
Bullet Bills	10	Bullet Bill projectiles
Floating Text	20	Score/combo text popups

16.2 Spatial Hashing (Collision Broadphase)

Property	Value
Cell Size	64 × 64 pixels (2 × tile size)
Rebuild	Every physics frame
Benefit	Only check collisions between entities in same or adjacent cells
Complexity	O(n) average vs. O(n ²) brute force

16.3 Config-Driven Entity Definitions

All entity properties defined in `entities.json`:

```

{
  "goomba": {
    "speed": 60,
    "health": 1,
    "score": 100,
    "stompable": true,
    "fireballKillable": true,
    "strategy": "patrol",
    "animation": { "walk": [0,1], "squish": [2], "frameDuration": 0.3 }
  }
}

```

Factory reads from this config, eliminating hard-coded entity values.

16.4 Replay System

Property	Value
Recording	Captures all input commands per frame
Storage	Binary file with frame-indexed command stream
Playback	Deterministic replay using same RNG seed + commands
File Size	~1KB per minute of gameplay
Use	Speed run verification, ghost playback

16.5 Debug Console

Property	Value
Toggle	~ (tilde) key
Commands	spawn <entity> <x> <y> , setlevel <n> , god , noclip , give <item> , kill_all , tp <x> <y> , speed <multiplier> , stats
Pattern	Command Pattern — text is parsed into ICommand objects
Autocomplete	Tab-completion for command names

17. Accessibility [NEW in v2.0]

17.1 Difficulty Modes

Mode	Lives	Enemy Speed	Checkpoints	Items
Easy	5	×0.75	Every 50 tiles	More frequent
Normal	3	×1.0	Mid-level (1 per level)	Standard
Hard	1	×1.25	None	Fewer items

Implemented via `DifficultyStrategy` that modifies game constants at level load.

17.2 Colorblind Mode

Mode	Adjustment
Deuteranopia	Green→Blue palette swap for enemies/items
Protanopia	Red→Yellow palette swap
Tritanopia	Blue→Pink palette swap

Toggle in Options menu. Applies shader/palette swap.

17.3 Audio Navigation Cues

- Menu selection: click sound on highlight change
 - Menu confirm: confirm chime
 - Menu back: back chime
 - Invalid action: error buzz
-

18. Meta-Game & Progression [NEW in v2.0]

18.1 Achievement System

Achievement	Condition	Icon
First Stomp	Defeat first enemy by stomping	Boot
Coin Collector	Collect 100 coins in a single run	Gold coin
Speed Demon	Complete any level in under 120 seconds	Clock
Untouchable	Complete any level without taking damage	Shield
Combo King	Achieve an ×8 combo	Star burst
Shell Shocker	Defeat 3 enemies with a single shell kick	Shell
Dragon Slayer	Defeat Bowser	Crown
Star Hoarder	Collect all 9 star coins	Star
No Deaths	Complete all 3 levels without dying	Heart
Secret Finder	Find all hidden blocks	Magnifying glass

- Toast notification on unlock (slides in from top-right, 3s display).
- Achievements viewable from Main Menu.
- Persisted across sessions.

18.2 Unlockable Characters

Character	Unlock Condition
Toad	Complete all 3 levels
Peach	Complete all 3 levels without dying

18.3 New Game+

Property	Value
Trigger	Complete all 3 levels
Changes	Levels are horizontally mirrored, enemies move 25% faster, fewer power-ups
Indicator	"★" prefix on world names

18.4 Daily Challenge

Property	Value
Seed	Current date (YYYYMMDD) → deterministic random seed
Generation	Procedurally generated level using the seed
Leaderboard	Local file (<code>daily_scores.json</code>) — score + time
Length	100 tiles wide (shorter than main levels)

19. Post-MVP Bonus Features

These features are **not in MVP scope** but are planned for post-MVP development if time allows.

19.1 Mario Maker (In-Game Level Editor)

- **Trigger:** Press F1 to pause and open editor overlay.
- **UI:** ImGui drag-and-drop: place tiles, enemies, items, coins.
- **Pattern:** Command Pattern for Undo/Redo.
- **Serialization:** Export/import levels as JSON.
- **Academic Value:** Nails the "Save/Load / Serialization" bonus requirement.

19.2 Time Manipulation ("Braid" Mechanic)

- **Trigger:** Hold Shift to rewind time.
- **Implementation:** Store game state snapshots in a circular buffer (~5 seconds = 300 frames).
- **Pattern:** Memento Pattern.
- **Academic Value:** Demonstrates advanced data structures (circular buffer).

19.3 Shadow Mario (Rival AI)

- **Feature:** Dark Mario that mimics player inputs with a 3-second delay.
- **Implementation:** Record player input history, replay through AI controller.
- **Pattern:** State Pattern + Strategy Pattern for pathing.
- **Academic Value:** Goes beyond basic proximity AI.

19.4 Dynamic Lighting & Weather (SFML Shaders)

- **Feature:** Day/night cycle, underground darkness with player-centered light radius, rain effects.
- **Implementation:** `sf::Shader` (GLSL fragment shaders).
- **Academic Value:** Graphics programming, unique visual presentation.

19.5 Procedural Level Generation (Infinite Mario)

- **Feature:** "Endless Mode" with dynamically generated terrain.
 - **Implementation:** Chunk-based generation with difficulty scaling (Perlin noise or rule-based).
 - **Academic Value:** Algorithmic thinking, procedural generation.
-

20. Rubric Alignment

Requirement	Points	Our Implementation
Player Inputs, Movement, Collision	20	AABB + spatial hash, Command pattern input, 10 movement mechanics (walk, run, jump, wall jump, ground pound, crouch, slide, swim, climb, combo), 2-player support
Enemy Behavior	10	13 enemy types + 3 color variants with 7+ AI strategies, mid-boss, final boss with phases
Power-Ups and Items	10	12 item types, 7 player states, P-Switch tile transformation, POW broadcast
3 Level Completion	15	3 themed levels + bonus rooms + world map + vertical/autoscroll sections + star coins
Sounds	10	17+ SFX, per-level BGM, dynamic music layers, surface footsteps, combo SFX, volume controls
OOP Design	10	Deep 4-level hierarchy, 25+ concrete classes, SOLID principles, smart pointers, Template Method
5 Design Patterns	25	10+ patterns: Factory, Singleton, State, Observer, Strategy, Command, Decorator, Memento, Object Pool, Template Method
AI	5	7+ strategies, Lakitu spawner, HammerBro arcs, Thwomp state machine, Boo proximity, Boss phases, config-driven variants
Multiple Players	5	4 characters (Mario, Luigi, Toad, Peach), character select + hotkey switch, unlockable progression, 2P versus
Total	110/115	(no 3D = -5, compensated by GLSL shader effects)